

VALLEY, TX, USA

WMO: 722505

Lat: 26.228N

Lon: 97.654W

Elev: 34

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.0	40.7	22.6	17.0	53.2	27.7	21.7	50.4	32.2	76.3	28.6	73.3	12.0	340	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.0	98.9	77.6	97.3	77.7	95.9	77.6	81.0	89.4	80.4	88.8	79.9	88.4	16.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.4	152.7	82.5	78.9	150.4	82.4	78.2	146.5	82.3	44.5	89.3	43.9	88.8	43.3	87.7	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	25.0	22.8	32.3	102.1	3.8	2.6	29.5	103.9	27.3	105.5	25.1	106.9	22.3	108.8
			29.5	82.9	3.2	1.6	27.2	84.0	25.4	84.9	23.6	85.8	21.3	87.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.8	60.9	64.9	69.6	75.6	81.1	84.5	85.5	85.9	82.1	76.3	68.4	62.2
	DBStd	11.05	9.17	9.33	7.94	6.51	4.64	2.76	2.49	2.59	3.68	6.31	8.14	9.11
	HDD50	46	20	9	2	0	0	0	0	0	0	0	3	12
	HDD65	586	187	110	49	8	0	0	0	0	0	7	60	165
	CDD50	9095	359	425	608	768	963	1036	1099	1114	963	816	553	391
	CDD65	4161	61	106	190	326	498	586	634	649	513	358	161	79
	CDH74	43679	376	676	1295	2823	5229	6992	7845	8041	5325	3337	1227	513
	CDH80	19989	52	164	378	1068	2305	3516	4099	4252	2410	1321	336	88
(14) Wind	WSAvg	11.4	11.4	12.5	12.9	13.3	12.9	11.6	11.5	10.4	8.6	9.7	10.4	11.0
(15) Precipitation	PrecAvg	27.6	1.7	1.4	0.9	1.6	3.0	2.9	2.0	3.1	5.7	2.5	1.5	1.5
	PrecMax	45.3	4.6	7.3	3.9	10.4	12.0	7.5	8.7	9.0	17.7	6.3	4.3	4.3
	PrecMin	14.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.8	0.2	0.0	0.2
	PrecStd	7.1	1.4	1.4	1.1	2.2	2.6	2.0	2.1	2.3	3.7	1.7	1.3	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.2	88.4	91.5	95.3	97.3	99.4	99.7	100.3	99.2	94.0	89.6	85.4
		MCWB	68.4	69.9	72.0	74.8	78.1	78.5	77.1	77.7	76.6	76.6	74.8	71.5
	2%	DB	81.2	84.4	87.2	91.6	94.5	97.0	98.3	98.9	95.4	91.4	86.6	82.4
		MCWB	68.8	70.7	71.9	74.1	77.4	78.1	77.3	78.0	77.6	76.2	73.1	71.0
	5%	DB	78.7	81.7	84.1	89.4	92.7	95.2	96.9	97.3	93.1	89.8	84.1	80.1
		MCWB	67.1	69.8	71.1	73.9	76.6	77.7	77.5	77.9	77.6	75.7	72.1	69.8
	10%	DB	75.0	78.7	81.4	86.4	90.6	93.3	94.9	95.5	91.0	87.6	81.4	76.6
		MCWB	65.8	68.0	69.4	72.8	76.2	77.6	77.7	77.8	77.3	75.1	70.8	68.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.8	74.2	75.4	79.0	80.7	81.9	81.3	81.4	81.4	80.5	77.5	75.0
		MCDB	79.0	82.4	85.2	88.3	91.2	91.6	88.4	90.8	88.2	86.2	85.1	81.0
	2%	WB	71.1	72.8	74.0	77.2	79.5	80.9	80.3	80.4	80.5	79.3	75.8	73.4
		MCDB	77.1	79.8	82.4	86.9	89.7	89.6	88.2	89.9	87.7	84.9	82.2	78.7
	5%	WB	69.8	71.4	73.0	76.0	78.5	80.2	79.7	80.0	79.9	78.3	74.6	72.1
		MCDB	75.1	77.6	80.2	84.8	87.7	88.6	88.1	89.5	87.4	84.9	80.5	77.3
	10%	WB	68.3	70.4	71.8	75.0	77.8	79.5	79.1	79.3	79.1	77.3	73.2	70.5
		MCDB	73.6	75.9	77.9	82.9	86.3	87.5	87.8	88.9	86.7	84.5	78.7	75.4
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	19.8	19.6	19.5	18.5	18.8	19.4	20.0	17.9	19.9	20.1	20.0
		MCDBR	22.2	21.3	20.6	20.9	19.6	20.4	21.5	21.7	20.3	20.8	21.1	21.4
		MCWBR	11.7	9.4	8.0	7.2	5.7	5.0	4.8	4.9	6.1	7.7	9.5	11.8
	5% WB	MCDBR	19.0	18.0	17.6	17.9	17.2	17.4	18.7	19.8	16.9	16.9	17.8	18.6
		MCWBR	11.2	9.0	8.1	7.1	5.8	5.3	5.2	5.4	6.1	7.7	9.6	11.9
(40) Clear-Sky Solar Irradiance	taub		0.361	0.375	0.398	0.438	0.449	0.426	0.445	0.431	0.438	0.390	0.376	0.364
	taud		2.468	2.418	2.353	2.250	2.252	2.378	2.334	2.379	2.325	2.457	2.452	2.461
	Ebn at Noon		282	286	285	274	270	274	269	273	268	278	274	275
	Edh at Noon		31	35	39	44	44	39	41	39	40	33	31	30
(44) All-Sky Solar Radiation	RadAvg		986	1185	1458	1732	1913	2088	2080	1988	1652	1446	1110	889
	RadStd		107	175	155	146	124	75	140	86	118	134	115	120

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189

Nomenclature: See separate page